

RCTX/RMAP Boundary: Shared Machine-Context Identity

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Abstract

The civic evidence package family needs a shared machine-context layer, but that layer should not absorb every use of geography. This paper documents the X.2 boundary between RCTX and RMAP. RCTX owns canonical unit identity, context hashes, graph identity, source hashes, aligned attributes, and exact rational crosswalks. RMAP is reserved for rendered and cartographic presentation: styling, layer composition, projections for display, tiles, images, and public map hashes. The implemented surface is deliberately narrower than the concept. The existing RPLAN `.rctx` implementation remains RCTX-compatible phase-1 infrastructure, while `rctx-core` exposes the shared package, crosswalk, graph-reference, source-reference, content-hash, and claim-boundary checks needed by sibling products. A standalone RMAP crate and full `rctx-io` loader remain deferred until a second production consumer requires them.

1 Introduction

RCTX is the shared machine-context contract for the civic evidence package family. Its job is to say which units exist, in which canonical order, under which context hash, with which graph identity, source hashes, aligned attributes, and crosswalks. That contract lets RPLAN, RCOUNT, RHIST, and later packages refer to the same base dimension without making each package invent a private geography model.

The boundary is intentionally conservative. RPLAN already has an implemented `.rctx` artifact in `rplan-core` and `rplan-io`. X.2 does not replace that path. It treats the current `rplan_core::RplanContext` shape as RCTX-compatible phase-1 infrastructure and keeps full context construction where it already works. The shared extraction is limited to the parts that sibling products already need: context unit indexes, source references, graph records, exact rational crosswalks, package hashing, and claim-boundary checks.

RMAP is a separate owner concept. It is reserved for rendered or cartographic presentation: maps, layers, styling, labels, projections for display, tiles, images, and map-image hashes. RMAP may reference an RCTX context hash, but RCTX must remain useful without a rendered map. This is the central X.2 claim: machine context should be shared, while cartographic presentation should not be mistaken for canonical unit identity.

2 Ownership Contract

The RCTX boundary spec records the dependency rule: RPLAN, RCOUNT, RHIST, and RMAP may depend on RCTX concepts, but RCTX must not depend on those domain packages [Della-Libera, 2026b]. The civic evidence package family spec uses the same split: RCTX is the canonical machine context, and RMAP is the public map presentation layer.

Owner	Contract
RCTX	Stable unit ids, unit namespace, canonical order, context hash, unit-universe hash, graph identity, source hashes, aligned non-domain attributes, and crosswalks.
RMAP	Rendered maps, styling, symbols, colors, labels, layer composition, display projections, raster/vector tiles, and public map image hashes.
RPLAN	District assignments, plan metadata, plan audit certificates, and the current <code>.rctx</code> read/write path until another full-context producer needs extraction.
RCOUNT	Election totals, CVRs, canvass records, ballot manifests, audit evidence, and references to RCTX context or crosswalk hashes through count-package metadata.
RHIST	Unit lineage across cycles: splits, merges, renames, boundary edits, and historical cross-cycle relationships.

The implemented `rctx-core` record shape follows that ownership line. `RctxPackage` contains a `RctxManifest`, a source index, context unit indexes, graph records, crosswalk records, and a claim boundary. `RctxManifest` names the RCTX version, package id, jurisdiction, producer, creation time, and `package_content_hash`. A `ContextUnitIndex` binds a `context_hash` to ordered `unit_ids`. A `GraphRecord` binds graph identity to a context hash, graph hash, and source references. A `CrosswalkRecord` binds source and target contexts, endpoint units, an exact rational weight, a weight kind, an exhaustiveness flag, and source references.

The existing RPLAN context maps directly onto the shared boundary:

RCTX concept	Existing <code>rplan_core::RplanContext</code> field
Schema version	<code>rctx_version</code>
Context digest	<code>context_hash</code>
Canonical unit universe	<code>units: PlanUnitIndex</code>
Unit kind, state, year, order, source id	<code>PlanUnitIndex</code> fields
Unit-universe hash	<code>unit_universe_hash</code>
Adjacency graph	<code>graph: Option<UnitGraph></code>
Aligned population vector	<code>populations: Option<Vec<i64>></code>
Aligned subdivision ids	<code>subdivisions: Option<SubdivisionContext></code>
Aligned demographic vectors	<code>demographics: Option<DemographicContext></code>
Geometry identity references	<code>geometry: Option<GeometryContext></code>
Source registry	<code>source_hashes: SourceHashes</code>

That mapping is why X.2 does not move the full `.rctx` implementation. The as-built RPLAN context already carries the fields needed by the shared contract; the extraction point is the cross-product verification primitive, not a rewrite of working plan context I/O.

3 Implemented Surface

The real shared crate is `crates/rctx-core`. Its public surface includes `RctxPackage`, `RctxManifest`, `RctxSourceIndexEntry`, `ContextUnitIndex`, `GraphRecord`, `CrosswalkRecord`, `RationalWeight`, `CrosswalkWeightKind`, `CrosswalkVerificationInput`, `ClaimBoundary`, and `VerificationReport`. The consumer-facing functions include `verify_package`, `verify_crosswalk_in`, `package_content_hash`, `crosswalk_record_hash`, `crosswalk_set_hash`, and `is_sha256_hash`.

`verify_package` first checks the manifest, derives the source set, verifies crosswalk input, verifies graph context and source references, and then verifies the claim boundary. Its reports expose named checks such as `manifest_package_hash`, `context_unit_index`, `source_index`, `crosswalk_unit_refs`, `crosswalk_weight_sum`, `crosswalk_source_refs`, `graph_context_refs`, `graph_source_refs`, and `claim_boundary_present`.

Surface	Implemented check
<code>ContextUnitIndex</code>	Context hashes must be valid <code>sha256</code> : values; unit lists must be non-empty and duplicate-free.
<code>RctxSourceIndexEntry</code>	Source ids must be unique; source hashes must be valid <code>sha256</code> : values.
<code>GraphRecord</code>	Graph records must reference known contexts, carry valid graph hashes, and reference known sources.
<code>CrosswalkRecord</code>	Endpoint contexts and units must exist; weights must be non-negative rationals with positive denominators; duplicate rows are rejected; exhaustive groups must sum exactly to one.
<code>ClaimBoundary</code>	The boundary package id must match the manifest, and both <code>proves</code> and <code>does_not_prove</code> must be present.

The crate is already shared with sibling products. RHIST depends on `rctx-core` for crosswalk verification, RCOUNT records RCTX references through `normalized/rctx-refs.ndjson`, and district aggregation code can bind RCOUNT reporting units, an RPLAN assignment, an RCTX context hash, and RCTX crosswalk records by hash. Those uses do not make RCOUNT or RHIST owners of the base context.

The synthetic fixture in `rctx-core` is small but representative. It constructs `syn-rctx-10-shared-cont` with one source-index entry, one context unit index, one graph record, two identity crosswalk rows, and a claim boundary that proves only declared unit uniqueness, graph/crosswalk references, and exhaustive crosswalk sums. It explicitly does not prove legal geography, source completeness, vote totals, district assignments, or rendered maps.

Two surfaces are deliberately deferred. RMAP is not a built crate in this repository; it is a reserved ownership boundary for future rendered map products. A standalone `rctx-io` loader is also deferred until a second consumer needs independent package-directory I/O rather than the current helper and fixture shape.

4 Boundaries

RCTX is a bridge, not an ownership transfer. It provides a shared base dimension that domain packages can reference. It does not own plan assignments, vote totals, ballot records,

audit samples, lineage events, certification claims, court records, or rendered maps. Those facts remain with RPLAN, RCOUNT, RHIST, RCERT, RCASE, or RMAP according to the package-family boundary.

The claim boundary is encoded in code as well as prose. `ClaimBoundary` requires a package id, a `proves` list, a `does_not_prove` list, and optional caveats. `verify_package` rejects a package whose claim boundary names the wrong package or omits either side of the claim. That shape is modest, but it prevents a shared context package from silently becoming a legal or cartographic certification.

RMAP is reserved, not implemented. The specs assign RMAP rendered maps, styling, symbols, labels, layer composition, display projections, tile products, and map-image hashes. The repository does not currently contain a standalone RMAP crate, and this paper does not claim one exists. The correct near-term relationship is that a future RMAP product may reference an `RCTX context_hash`; it should not invent canonical unit identity on its own.

The split decision is also explicit. The plan and boundary spec say not to move full `.rctx` read/write out of RPLAN until a second production consumer needs direct full-context construction or validation. `rctx-core` exists because crosswalks, source refs, context unit indexes, graph records, content hashes, and claim boundaries already have cross-product pressure. That is a different decision from splitting all RPLAN context I/O.

5 Conclusion

X.2 fixes the RCTX/RMAP boundary in the direction the code now supports. RCTX is the shared machine-context identity layer: unit identity, canonical order, context hashes, graph identity, source hashes, aligned attributes, and crosswalks. RMAP is the separate owner of rendered and cartographic presentation, and remains reserved rather than implemented.

The implemented surface is narrower and more useful than a premature full split. `rctx-core` gives sibling products a shared package and verifier surface for context unit indexes, source references, graph records, exact rational crosswalks, package content hashes, and explicit claim boundaries. RPLAN keeps the current `.rctx` implementation until another full-context producer needs it. That preserves the working plan pipeline while giving RCOUNT, RHIST, and later packages a common base dimension to reference.

References

- Gio Della-Libera. Rcount overview: Reproducible election count packages. Technical report, Apportionment Research, 2026a.
- Gio Della-Libera. Rctx boundary: Shared reproducible machine context. Technical report, Apportionment Research, 2026b.